Shock-diminished paleomagnetic remanence at the Charlevoix impact structure, Quebec[1]

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The magnetic characteristics of shocked and unshocked anorthosites of the Charlevoix impact structure have been reexamined to assess the effects of high thermal and alternating field (af) cleaning treatments, and to evaluate any shock-induced features of the remanence in terms of mineralogy of the shocked rocks.

From 16 sites, 197 specimens were subjected to stepwise thermal treatments to 710°C, and a further 46 to incremental af cleaning to 290 mT. The 10 sites from the unshocked St-Urbain anorthosite carry a magnetization with unblocking temperatures (T_{UB}) generally between 600–625°C, with a significant fraction of remanence with $T_{UB} > 670$°C. Similarly, most of the remanence has a resistive coercive force (rcf) of 100–175 mT, with a significant fraction exceeding 290 mT. The remanence is carried by two exsolution phases of titano-hematite, with the later and more hematite-rich lamellae having the higher T_{UB} and rcf. Although only one direction exists, at two sites where intensities are somewhat lower and some natural remanent magnetization (NRM) directions scattered, treatment reveals a dual polarity remanence. The pole (154°E, 02°S) obtained from these unshocked St-Urbain anorthosite sites falls on a well established segment of the late Precambrian apparent polar path of suitable age (≈950 Ma). Significant results could not be obtained from three additional sites in the marginal zone due to their ready acquisition of a viscous magnetization following thermal treatments.

Data from two sites which display no shock effects and lie near the margin of the central uplift, and from one highly shocked sample from the crater centre show some possible effects of the shock event. All three have intensities substantially lower than in the St-Urbain sites, which is interpreted in the case of at least one of these sites as a shock-diminished remanence. A puzzling feature of the highly shocked samples is that the phase with a texture reminiscent of the ilmenite–hematite exsolution in the unshocked anorthosites, contains no iron. The remaining remanence, whose direction is parallel with that of the unshocked sites, is interpreted as pre-crater rather than shock-produced. Although many instances have been reported of a new shock-induced remanence imparted to rocks of equivalent and even lesser shock grade than those examined at Charlevoix, in all cases the magnetic carrier seems to have been magnetite. The lack of a shock-induced remanence at Charlevoix is attributed to the high T_{UB} and rcf of hematite in these rocks.

Les caractéristiques magnétiques d'anorthosites près du point d'impact de la structure de Charlevoix sont le sujet de cette étude dont le but est d'établir l'effet d'un choc sur la remanence. Nous donnons un compte-rendu de la minéralogie et du comportement sous traitements thermiques et par champ alternatif d'anorthosites qui furent affectées par le choc et d'autres qui demeurèrent inaffectées.

De 16 sites, 243 échantillons furent soumis à des traitements par étapes soit thermiques allant jusqu'à 710°C (197 d'entre eux), soit par champ alternatif allant jusqu'à 290 mT (46). Aux neuf sites qui échappèrent au choc, nous retrouvons une seule aimantation dont le gros des températures de déblocage (T_{DB}) se situent entre 600–625°C et une partie, tout de même appréciable, qui a des $T_{DB} > 670$°C. Pareillement, le plus gros de l'aimantation a des forces coercitives résistantes (fcr) de 100–175 mT et une partie a des fcr qui résistent à un champ alternatif de 290 mT. L'aimantation est contenue dans deux phases d'exsolution de titanohématites; la dernière avec ses lamelles très riches en hématite contient tout probablement l'aimantation à T_{DB} et fcr les plus hautes. A deux sites, les intensités sont faibles et les directions initiales sont plutôt dispersées; les traitements nous revèlent que ceci est dû au fait qu'une partie de l'aimantation est dans un sens et le restant dans le sens opposé pour former une aimantation à polarité double. Le pôle (154°E, 02°S) que l'on obtient de ces sites tombe sur une portion du cheminement du pôle par

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rapport à l'Amérique du Nord à la calibration 950 Ma, ce qui s'accorde bien avec l'âge probable de l'anorthosite de St-Urbain. Nous n'avons tiré aucun résultat significatif de trois autres sites provenant de la zone marginale, dû à l'acquisition d'une aimantation visqueuse après chaque traitement thermique. Nous avons aussi étudié deux sites plus près du point d'impact qui visuellement n'apparaissent pas avoir été affectés par le choc en plus d'un autre site situé au point d'impact. Les échantillons des trois sites ont des intensités beaucoup plus faibles que celles retrouvées aux sites de St-Urbain et notre interprétation est que cette réduction de rémanence est attribuable au choc. Une énigme que nous rencontrons dans les échantillons du point d'impact est que les minéraux opaques qui ont une texture d'exsolution semblable à celle de l'ilménite–hématite des autres sites ne semblent contenir aucune trace de fer. L'aimantation restante dont la direction s'accorde bien avec celle des autres sites est interprétée comme prédatant la formation du cratère plutôt que produite par le choc. Quoique souvent on ait rapporté que des roches qui furent soumises à un choc moindre que celles de Charlevoix avaient acquis une rémanence induite par le choc, il faut remarquer que dans tous les cas, le porteur d'aimantation était de la magnétite. Nous attribuons donc le manque de rémanence induite à Charlevoix au fait que l'aimantation de ces roches est portée par de l'hématite qui a de très hautes T_{DB} et fcr.

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Introduction

The effects of dynamic stress on magnetic remanence have been directly demonstrated in experiments, and described from several natural situations. Shapiro and Ivanov (1967), Nagata (1971), and Pohl *et al.* (1975) showed that natural remanence of magnetite in basalt is destroyed by shocking samples in a zero-field, and also that a shock remanence is induced by shocking demagnetized samples in a magnetic field. In nuclear explosion experiments, shocked basalts from the Danny Boy event possess an extremely stable shock induced remanence (Wasilewski 1973). Measurements on granodiorite from the Hardhat nuclear crater, however, and on tuff from the Rainier event are inconclusive in terms of detecting the effect of shock on remanence intensity or direction (Hargraves and Perkins 1969), although bulk susceptibility may have been reduced. In natural events, a thermally stable remanence with direction consistent with the contemporary Earth's field has been acquired by target rocks melted by the impact at several terrestrial hypervelocity meteorite craters. Notable in this regard are the suevite deposits of the Ries crater (Angenheister and Pohl 1964) and the thick sheets of impact melt at the Manicouagan (Larochelle and Currie 1967) and Mistastin (Currie and Larochelle 1969) impact sites. The effects reported for less strongly shocked crater rocks, however, are not as consistent. Cisowski (1975) noted a shock demagnetization and a shock remagnetization in basalts from the ejecta and rim of the Lonar crater, and implied that these effects were directly proportional to the shock grade of the samples. Samples from the rim of Barringer Meteor Crater, on the other hand, show no effects of shock on their remanence (Hargraves and Perkins 1969). Mafic dykes and volcanics near the margin of the central uplift at the Slate Islands impact structure possess a secondary component of magnetization, as do dykes of impact breccias which cut these rocks (Halls 1975). The directions of the two differ, however, and Halls and Stesky (1978) believe the secondary component of the volcanics is shock produced, whereas that of the breccia is a TRM (thermoremanent magnetization) acquired after frictional heating during breccia injection. Hargraves and Roy (1974) discovered reduced remanence intensity in shocked anorthosites from the central uplift of the Charlevoix crater, compared with intensities in anorthosite lying near the margin of the structure. The remanence direction of the shocked samples, although different from that in the unshocked material, varied considerably and Hargraves and Roy (1974) could not determine whether this direction was shock induced, a rotation of rocks bearing the pre-impact direction, or a combination of both.

In their study, Hargraves and Roy (1974) demagnetized samples in alternating fields to a maximum of 50 mT (1 mT = 10 Oe). For the vast majority of sites, where a consistent NRM (natural remanent magnetization) direction was obtained, no significant change in either direction or intensity was revealed by af demagnetization. The present study was undertaken, therefore, in the belief that a more complete, and perhaps clearer, magnetization history of the Charlevoix anorthosites could result through thermal demagnetization in the range of post-shock temperatures experienced in the impact event, and beyond to the Curie temperatures of the magnetic carriers. Treatment in alternating fields considerably higher than those employed by Hargraves and Roy (1974) could also aid in interpretation. In addition, if the changes in remanence intensity and direction of the highly shocked anortho-

TABLE 1. Analysis of directions before (NRM) and after thermal and alternating field treatments of

	NRM						T_{UB}(°C)		rcf (mT)		Thermal			
Site	N_1	Int. (mA m^{-1})	SD	$D°, I°$	κ	α_{95}	Range	Max.	Range	Max.	N_2	$D°, I°$	κ	α_{95}
1	6	137	19	128, +50	180	5	600–625°	>670°	100–175	>290	6	133, +48	145	6
2	8	158	34	214, +79	42	9	600–625°	>685°	150–250	>290	8	204, +72	394	3
3	9	180	40	093, +01	94	5	575–625°	>685°	125–225	>290	7	101, +05	114	6
4	8	173	54	013, +78	61	7	575–625°	>685°	125–200	>290	8	349, +80	66	7
5	8	4687	2624	246, +44	2	60								
6	8	942	476	106, +37	2	50								
7	7	84	52	200, +82	6	28								
8	11	102	43	096, +29	2	59	600–625°	>670°	100–200	250	6	109, +37	67	8
9	5	73	22	139, +71	475	4	600–625°	>670°	100–225	>290	5	138, +64	124	7
10	8	158	32	113, +33	83	6	575–650°	>670°	100–200	>290	8	129, +40	24	11
11	15	9.3	6.7	321, −11	5	20	530–650°	650°	0–150	150	14	292, −64	24	8
12	8	1.1	0.5	185, −37	19	13								
13	7	6.4	5.5	222, +22	13	17	100–650°	>650°	20–175	175	5	116, +49	9	15
14*a*	8	115	54	143, +24	10	19	100–650°	670°	100–200	>290	6	122, +21	19	16
14*b*	6	13	7	227, +41	4	38	500–670°	670°		>290	4	UN	—	—
15	9	27	13	065, +65	6	23	100–625°	>670°	50–175	275	8	143, +50	72	6
16	8	445	67	116, +53	212	4	575–625°	>685°	100–150	>290	8	118, +54	197	4

NOTES: For definition of N_1, N_2, N_3, and N_4 see text. The site-mean intensity (Int.) of NRMs is given together with the standard deviation (SD). D and I are the mean declination and inclination respectively. R is the resultant vector; κ is the precision estimate; α_{95} is the circle of confidence at $P = 0.05$. Most of the remanence decays in the heating (T_{UB}) and alternating field (rcf) ranges indicated. Some of the remanence, however, resists the maximum temperature and alternating fields as indicated. The treatments after which the D and I were computed are given. Statistics of site means are given in Table 2 following the key system shown. UN means unresolved dual polarity remanences (see text).

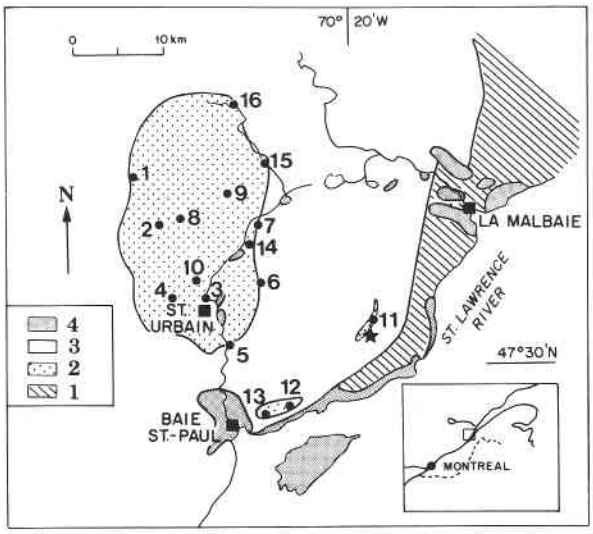


FIG. 1. Location of the Charlevoix, P.Q., impact structure (inset) and generalized geology (after Hargraves and Roy 1974). Geological units: (1) granitic gneiss, (2) anorthosite, (3) charnockitic gneiss, (4) Paleozoic sediments. The 16 sites sampled for paleomagnetism measurements are shown. The star marks the crater's centre, Mont des Eboulements.

sites are shock-induced, evidence might be obtained from petrographic examination, an aspect on which Hargraves and Roy (1974) did not report.

Location and Sampling

The Charlevoix impact structure was excavated largely in Grenville age charnockitic and granitic gneisses overlain by a relatively thin cover of Ordovician limestones (Fig. 1). Two anorthosite bodies lie in the western part of the affected area, the larger St-Urbain pluton being intersected by the downfaulted peripheral trough of the impact structure. A relatively narrow marginal zone of gabbroic anorthosite or diorite surrounds the anorthosite. Small isolated patches of anorthosite lie within the central uplift of the impact crater, approximately 10–14 km from the St-Urbain body. One of these outcrops occurs within the group of hills forming the central peak and, apart from a small remnant of impact melt, represents the highest grade of shock metamorphism preserved in outcrop (Robertson 1975). The St-Urbain anorthosites are cut by ilmenite-bearing dykes. These are dated at 890 Ma (Rose 1961) and believed to be only slightly younger than the anorthosite itself, whereas the impact event is dated at 360 ± 25 Ma based on the impact melt rocks (Rondot 1971).

The St-Urbain anorthosite is characterized by low magnetic field intensity surrounded by a belt of strong positive aeromagnetic anomalies coincident with the more mafic marginal phases. The charnockitic and granitic gneisses are of intermediate magnetic intensity (Rose 1961).

For the paleomagnetic study 16 sites were sampled; 13 within the St-Urbain anorthosite and marginal zone, three from the central uplift (Fig. 1).

the St-Urbain anorthosite and surrounding area which includes the Charlevoix impact structure

af						Combined											
Treatment	N_3	$D°, I°$	κ	α_{95}	Treatment	N_4	$D°, I°$	R	κ	α_{95}	Pole long (°E), Lat	Key for Table 2 statistics					
650°	1	135, +56	—	—	150	7	134, +49	6.95	138	5	148, 02°N	1	2	3	4	5	6
650°	1	212, +75	—	—	175	9	204, +72	8.98	402	3							
650°	6	094, +10	155	5	225	13	098, +08	12.88	101	4	191, 02°N		2		4		6
650°	1	290, +78			225	9	342, +80	8.87	66	6							
600°	7	104, +41	73	7	200	13	107, +39	12.83	71	5	173, 06°S	1	2	3	4	5	6
650°	1	106, +68	—	—	200	6	133, +65	5.95	91	7	141, 13°S	1	2	3	4	5	6
650°	1	115, +24	—	—	200	9	127, +39	8.67	24	11	158, 06°N	1	2	3	4	5	6
530–650°	1	322, −16	—	—	150	15	295, −62	14.24	18	9	153, 17°S			3	4	5	6
575–670°	1	UN	—	—		5	120, +49	4.84	24	16	158, 04°S			3	4	5	6
670°	4	132, +26	39	16	290	10	126, +23	9.63	24	10	164, 13°N			3	4		
	4	UN	—	—													
575–625°	1	129, +46	—	—	200	9	142, +50	8.89	71	6	142, 04°N	1	2	3	4	5	6
650°	1	106, +49	—	—	200	9	116, +53	8.95	175	4	159, 09°S	1	2	3	4	5	6

Sites 1, 11, 12, and 13 correspond, respectively, to sites 93, 92, 99, and 90 of Hargraves and Roy (1974). Samples from all sites were obtained by core-drilling and oriented by magnetic compass. Where suncompass measurements could also be taken (10 sites), orientations were in agreement. Based on the geology in the neighbourhood of the other six sites there is no reason to believe that orientations taken by magnetic compass alone contain appreciable error.

Methodology

Five to 15 cores (2.5 cm diameter) (N_1 in Table 1) per site were drilled in the field to yield a total of 137 cores, with one to four specimens (thickness = 2.2 cm) cut from each core. Stepwise demagnetization of 243 specimens was carried out in quasi-zero residual fields (<2 nT) (Roy *et al.* 1969). Of these, 197 were subjected to thermal and the remaining 46 to af treatment.

For each site, one or two pilot specimens were heated to 75°, then by increments of 100° from 100–500°, from 500–650° (or 655° in some runs) in increments of 25°, and then at 670, 685, and 710°C. The remaining specimens were treated following the same schedule from 500°C upwards, with a step also at 100 or 300°C for some sites.

Also for each site, one or two pilot specimens were af treated in incremental steps of 5 mT up to 15 mT, in steps of 10 mT from 30–70 mT, in steps of 25 mT between 100 and 275 mT, and then at 290 mT. The remaining specimens were af treated according to the above schedule from 100–290 mT, preceded by a treatment at 30 mT, and in some cases by additional treatments between 30 and 100 mT. The treatments were performed using apparatus described by Roy *et al.* (1972, 1973) and the measurements were made with Schonstedt SSM and DSM spinner magnetometers.

The results of demagnetization are compiled in Table 1. Each core is given unit weight. Hence, N_1 is the total number of cores treated per site. Thermal and af treatments sometimes yield different results; the two are reported separately as N_2 and N_3, respectively, and combined under N_4 ($= N_2 + N_3$). In instances where a core has provided specimens for both af and thermal treatments N_4 is larger than N_1. Information concerning the unblocking temperatures (T_{UB})[2] and the resistive coercive forces (rcf) of the high stability remanence is given in Table 1. Indicated are the temperature and af ranges where most of the remanence is removed. Generally, part of the remanence resists

[2]The T_{UB} is used to describe the temperature at which the remanence unblocks in laboratory experiments. For simple TRM the T_{UB} corresponds to the T_B (blocking temperature) as demonstrated by Thellier (1966). However, for remanences of other origins (depositional, chemical, or thermo-chemical) the T_{UB} does not necessarily correspond to the T_B, as the treatment mode most likely differs from the mode of acquisition. For example, hematite may be formed and magnetized at almost any temperature below its Curie point (680°C) and yet it may not be unblocked until treatment in the 600–700°C range. Since the origin of magnetization cannot be determined from laboratory thermal experiments alone, it appears preferable to distinguish between T_{UB} and T_B in order to avoid any implication regarding the origin of magnetization.

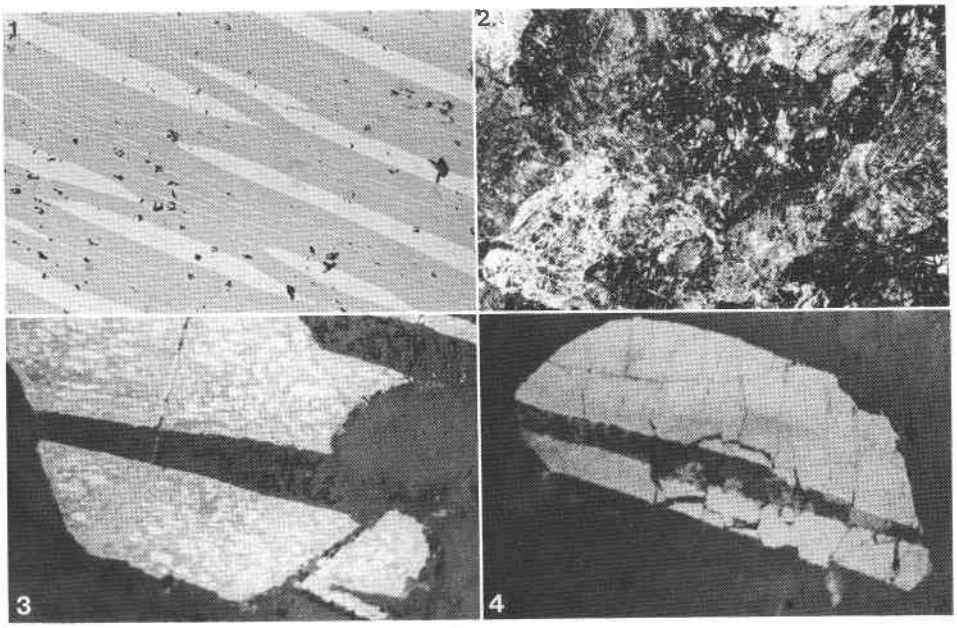


PLATE 1.

FIG. 1. Exsolution of titano-hematite (light gray) and ilmenite (darker gray) in unshocked St-Urbain anorthosite. Reflected light; width of field 0.25 mm.

FIG. 2. Incipient development of maskelynite in shocked anorthosite of site 11. Dark grains in centre of field, and darker patches in grain at lower left are isotropic or have highly reduced birefringence. Crossed polarizers; width of field 4.5 mm.

FIG. 3. Shocked "opaque" in anorthosite of site 11. Texture is somewhat similar to exsolution texture in Fig. 1. However, TiO_2 phase (lightest patches), sphene (dark patches), and a single phase with a combination of the above chemistries (intermediate shade) all lack iron. Reflected light; width of field 0.25 mm.

FIG. 4. As in Fig. 3 except that almost the entire grain comprises the phase with the combined TiO_2–$CaTiSiO_5$ chemistry. Reflected light; width of field 0.25 mm.

even more rigorous treatment as shown by the maximum temperatures and af treatments where remanence is still clearly observed. The results were compiled for a treatment step lower than the maximum (as indicated in Table 1) in order to avoid observations influenced by a low signal/noise ratio at these uppermost levels. At some sites the T_{UB} varied among specimens (site 15) or the intensity was too low (site 11) to obtain a meaningful reading from a single treatment step. In such cases a temperature range has been indicated instead of a single treatment step. Vector analysis has been employed to support our arguments, but no direction obtained in this manner was used to determine the pole position (Table 2).

Results

In terms of their location, petrography, and remanence characteristics the samples can be grouped into three categories: unshocked St-Urbain anorthosites ($N = 11$), the marginal-phase diorites ($N = 2$), and shocked anorthositic rocks of the central uplift ($N = 3$). Acquisition of a viscous magnetization after heating prevented satisfactory results for the marginal-phase diorites (sites 5 and 6) and a cataclastic anorthosite (site 7), also from the anorthosite margin. There will, therefore, be no further discussion of these sites other than their inclusion in Table 1.

Petrography

Two or more samples from each of the 16 sites were examined petrographically. Details of mineralogy and texture of the St-Urbain anorthosite are given by Mawdsley (1927) and Rose (1961). The metallic oxides, generally subrounded grains less than 0.5 mm in diameter, are an iron-rich ilmenite containing exsolution lamellae of titano-hematite (Pl. 1, fig. 1). Smaller lamellae of ilmenite in the hematite phase, plus smaller hematite lamellae in the ilmenite matrix, are believed to be a later stage of exsolution with compositions closer to the solid-solution end members (Carmichael 1961). Additional phases within the iron–titanium oxides are of minor abundance and remain unidentified. Plagioclase contains abundant, rod-like oriented inclusions (average: 50 μm × 5 μm) of rutile and iron oxides. The latter appear to be predominantly hematite although larger rods are two phases, probably members of the hematite–ilmenite series.

Site 14 lies on the eastern margin of the St-Urbain

pluton, but samples show petrographic and remanence differences from the anorthosites described above. The rocks exhibit cataclasis on a microscopic scale typical of many anorthosites. The opaque minerals are of two types: titano-hematite exsolution lamellae in an iron-rich ilmenite host, and larger grains (up to 1 mm) of a largely oxidized sulfide. Rod-like inclusions of hematite are generally rare within plagioclase grains.

The remaining sites (11, 12, and 13) lie within the central uplift, are isolated from the St-Urbain complex, and are located within small areas mapped as anorthosite (Rondot 1969; Hargraves and Roy 1974). Samples from site 12, however, are gabbroic anorthosite, whereas site 13 samples are diorites. The rocks from both these sites also contrast with the St-Urbain anorthosites and diorites in their opaques. Iron oxides, believed to be magnetite or titano-magnetite, are present only in trace amounts as small (0.2 mm), single-phase grains. No inclusions of iron oxides occur within plagioclase. Although both sites lie near the outer limit of shatter cone development (Robertson 1975), such shock phenomena were not observed in outcrop nor are microscopic shock features present.

The highly shocked rocks of site 11 are anorthosites, although distinctive differences exist from those of the St-Urbain body, some of which are shock-produced and others are probably original. In general, plagioclase has been shocked to the stage of incipient development of diaplectic glass (maskelynite), with original grain boundaries generally still discernible (Pl. 1, fig. 2). A labradorite composition, An_{60} (cf. An_{38-46} for St-Urbain anorthosite) was determined from polysynthetic twinning in rare grains which appear only weakly deformed. Opaques are small, and present in trace amounts only. Grains are reminiscent of the ilmenite–titano-hematite exsolution assemblage of the St-Urbain anorthosites, with elongate patches of a white-reflecting phase in a gray host (Pl. 1, fig. 3). Preliminary electron microprobe determinations indicate, however, that the white phase is not hematite but a titanium oxide, possibly rutile or anatase. The gray host has a composition close to sphene, and a third component (Pl. 1, fig. 4) appears to be a combination of the two chemistries but is a single phase. Significant in terms of the magnetic characteristics is the virtual lack of iron in all these phases, plus the absence of minute inclusions of magnetic phases within the shocked feldspars.

Remanence

St-Urbain Anorthosite

Samples of sites 1, 2, 3, 4, 9, 10, and 16 within the St-Urbain anorthosite carry a remanence with a very high T_{UB} and very high rcf. Although most of the remanence has T_{UB} and rcf between 600–625°C and 100–175 mT, respectively, a substantial part of that remanence has T_{UB} of 675°C or higher, and rcf which resists an af treatment as high as 290 mT (Fig. 2; Table 1). There is, therefore, little doubt that the carrier of remanence is hematite. It is probable that the large intensity remanence with T_{UB} in the 600–625°C range and the lesser remanence with $T_{UB} > 670$°C are carried by different phases. The latter is perhaps attributable to the late-stage, minute exsolution lamellae since the Curie temperature of this presumed hematite-rich phase is higher than that of the larger, Ti-rich lamellae (Nagata and Akimoto 1961). There is no significant change in direction after removal of the major component of remanence (600–625°C) and the index of precision (k) remains high (Table 1). Such characteristics indicate a single, normal polarity remanence free

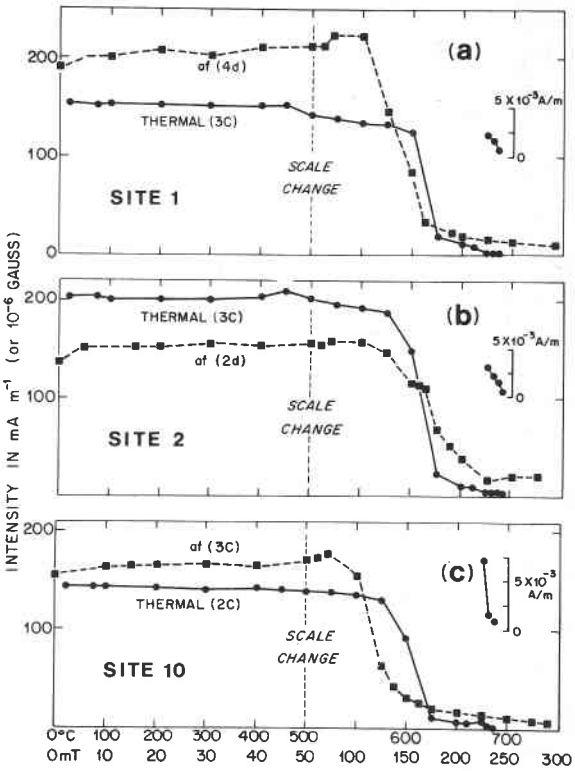


FIG. 2. Intensity decay curves of three thermally and three af treated specimens representative of the St-Urbain anorthosite. These rocks carry a univectorial remanence. Little directional change takes place throughout the treatment; the vector direction remains within 10° of arc about its mean. Although most of the remanence unblocks between 600 and 625°C, 5–10% remains after the 650° treatment and about 1% resists the 685 or 690° treatment. The 675–700° interval is shown as insets on an expanded scale. Likewise, although a large decay occurs in the 100–175 mT range, 5–10% resists af treatment as high as 290 mT.

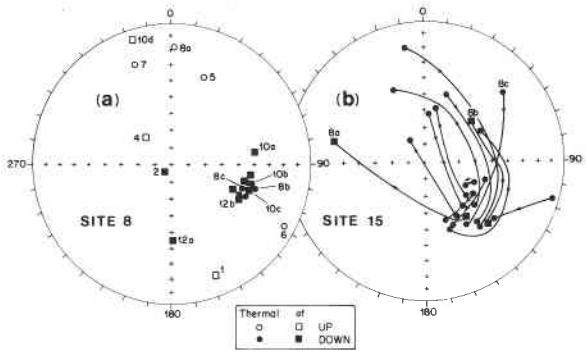


FIG. 3. Directions from the two sites of the St. Urbain anorthosite where the NRM directions are not tightly grouped: (*a*) shows only the NRM directions for site 8, (*b*) shows the directions before and after treatment for site 15. The symbols indicate the treatment to which the specimen was subsequently subjected. Specimens with a number followed by a letter were sliced from a single core. The similarity of the magnetic characteristics among specimens indicates that the scattered NRM directions represent the resultant vector of a dual polarity remanence. Equal-area projection.

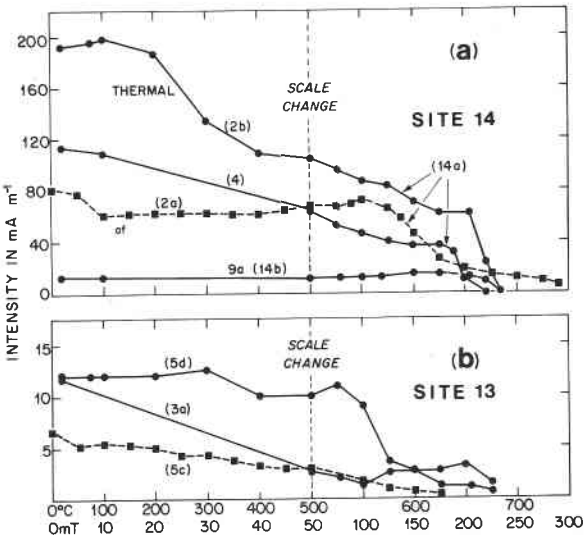


FIG. 4. Intensity decay curves of sites from: (*a*) the margin of the St-Urbain pluton and (*b*) the central uplift region.

from overprints and that the NRM was almost totally acquired during a single event, probably during cooling from above 700°C. This is supported by uniformity of NRM intensities among specimens within and between sites, suggesting a homogeneous distribution of magnetic particles throughout most of the pluton.

Somewhat more involved remanence characterizes the other three sites in the unshocked anorthosite. At sites 8 and 15 many NRM directions are scattered (Fig. 3). Cleaning treatments, however, show that a remanence of high T_{UB} and high rcf is present in all specimens and has a direction similar to the tightly grouped remanence directions of the other St-Urbain sites. The scatter of NRM directions at these two sites and the considerably lower intensities compared with the other St-Urbain anorthosites (Table 1) are interpreted in a following section as the result of an additional remanence of nearly opposite (reverse) direction.

Site 14 has been subdivided on the basis of magnetic intensities; the mean intensity of 14*b* being only 11% of that from site 14*a*. No petrographic distinction is evident between rocks of the two groups. The T_{UB} for site 14*a* are distributed over the 100–685°C range, although most of the rcf is limited to the 100–200 mT range (Fig. 4*a*). Nonetheless, an appreciable fraction of the remanence remains at 670°C and 290 mT and this has been used to calculate the site mean direction (Table 1). The magnetic characteristics of 14*b* are similar. Maximum T_{UB} and rcf are very high (> 670°C and > 290 mT), but most of the T_{UB} is restricted to the 500–670°C range. As at sites 8 and 15, the reduced intensity and more limited T_{UB} range at site 14*b* may also be the resultant of two remanences of opposite polarities and comparable stability spectra.

Shocked Rocks of the Central Uplift

The remanence characteristics of the shocked anorthositic rocks of the central uplift contrast with those of the St-Urbain anorthosites. At site 12, NRM intensities are weak (≈ 1.1 mA m^{-1}), in agreement with only trace amounts of magnetic minerals observed petrographically. After treatment the intensities fell below the sensitivity of the measuring instrument (1.0 mA m^{-1}) and no meaningful results could be obtained. The NRM intensities at site 13 are only slightly higher (6.4 mA m^{-1}), again reflecting the minor content of magnetic components. Unlike the St-Urbain sites, here the T_{UB} and the rcf are distributed over wide ranges, 100–650°C and 20–175 mT (Fig. 4*b*). In most specimens an appreciable fraction of the remanence remained in the 600–650°C range. No significant directional change was obtained from the single af treated specimen. The NRMs at site 11 are also weak and the directions scattered (Table 1). The T_{UB} are distributed mainly over the 300–650°C range and the rcf range is from 0–150 mT (Fig. 5*d*). Intensities fall to 1 mA m^{-1} in the 530–650°C range and the sole af treated specimen had an intensity of 1 mA m^{-1} at 150 mT. The high T_{UB} remanence is clearly reversely magnetized from the St-Urbain anorthosites, and projection on

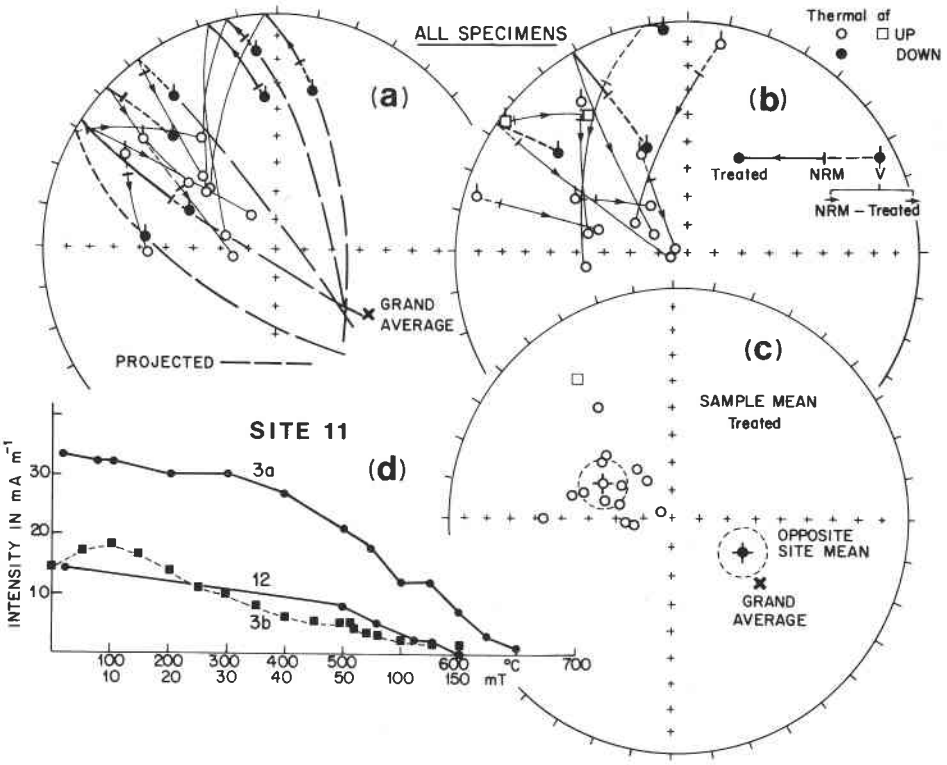


FIG. 5. Directions (*a*), (*b*), and (*c*) and intensity decay curves (*d*) from site 11 near the centre of the impact structure. The direction changes of all specimens are shown on stereonets (*a*) and (*b*) for greater clarity. Heavy lines/dashes are on the lower hemisphere, light lines/dashes are on the upper hemisphere (reverse polarity). The directions before (NRM) and after (Treated) treatment are shown. *V* was obtained by vector subtraction: the NRM vector minus the Treated vector. Projection through these points shows that most great circles pass close to the grand average obtained for this study (average 3 in Table 2), indicating that the NRMs also carry a small remanence of normal polarity. The projections (long dashes) of a few circles are shown in (*a*). The sample mean directions and the site mean direction with its α_{95} confidence circle are shown in (*c*). The agreement between the (opposite) site mean and the grand average mean indicates that the remanences were probably acquired contemporaneously.

great circles indicates that a remanence of lesser intensity and opposite (normal) polarity is probably present (Fig. 5*a*, *b*, *c*).

Evidence for Reversals

St-Urbain Anorthosite

All sites within the St-Urbain anorthosite clearly carry a normal polarity remanence of high T_{UB} and high rcf. At seven of these sites, only the normal polarity remanence *appears* to be present, as shown by the high κ in both NRM and treated results, and the small difference in site mean directions before and after treatment (Table 1). At other sites, some NRM directions are clearly different from the cleaned directions. The following descriptions of vector analysis show that this is most probably due to reverse polarity remanence (Figs. 6–8).

Projections of scattered NRM directions at site 8 all tend towards the reverse polarity (Figs. 6, 7). Core 8 is particularly interesting. The remanence of specimen 8*b* (Fig. 6*c*) remains in the normal polarity direction to 680°C, whereas for specimen 8*c* (Fig. 6*b*) the direction gradually shifts towards the reverse polarity from 625–670°C. The presence of two remanences of opposite polarity with overlapping T_{UB} spectra is well illustrated by specimen 8*a* (Fig. 6*c*). The direction moves towards the normal polarity at 550–600°C. At 650°C the remanence with reverse polarity is predominant and at 685–700°C the direction reverts to the normal polarity.

Remanences of opposite polarity could also be detected by af treatment. In core 10 of site 8 the three (*a*, *b*, and *c*) specimens cut from the top of the core apparently carry a single magnetization; the NRMs belong to the positive polarity group (Fig. 3) and very little directional change occurs during thermal and af treatments. The NRM of specimen 10*d*, however, is quite different and is northerly

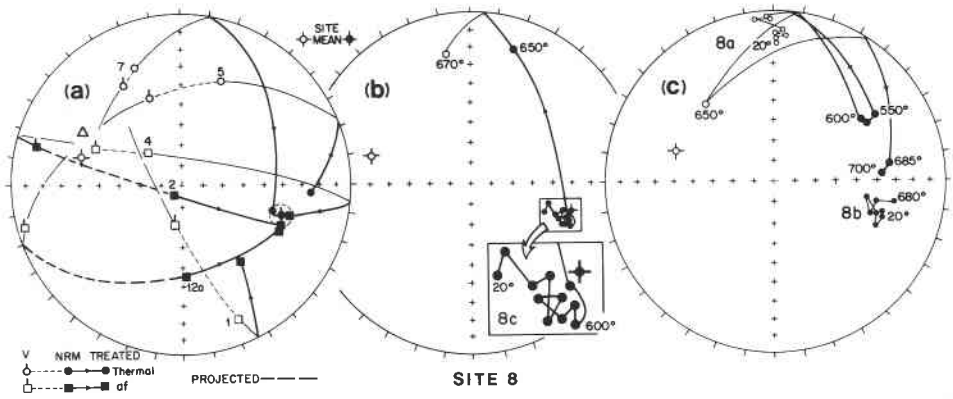


FIG. 6. Directions before (NRM) and after (Treated) treatment of all scattered NRMs of site 8 as shown in Fig. 3, except for specimen 10*d* shown in Fig. 7. *V* (vector subtraction) and projection as described for Fig. 5. The specimen number is shown opposite the NRM direction. Except for possibly specimen 1, the NRMs seem to be carrying both the normal and the reverse polarity remanences. The triangle is the mean of the subtracted vector directions for the six specimens shown in (*a*), specimen 8*a* in (*c*), and 10*d* in Fig. 7.

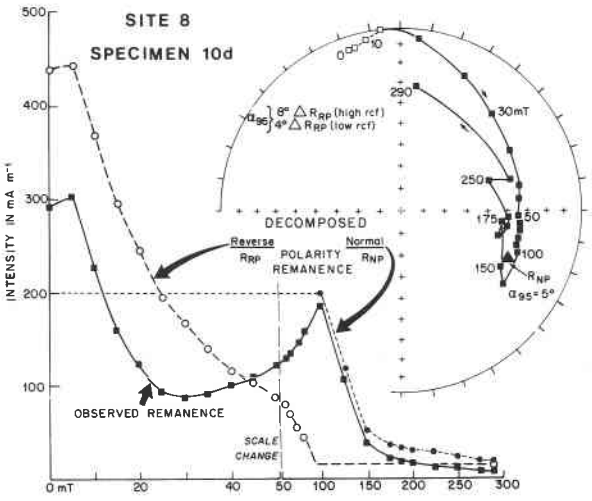


FIG. 7. Direction changes and intensity decay curve observed during af treatment up to 290 mT (squares). The dual polarity of the remanence is shown. The mean directions (triangles) of the normal (R_{NP}) and reverse (R_{RP}) polarity remanence have been obtained by averaging the directions of several vector subtractions calculated in the following manner. R_{RP} (high rcf) is the vectorial mean of the directions of the six vectors obtained by successively subtracting the 275–175 mT vectors from the 290 mT vector. R_{NP} is the mean of the 10 vectors obtained by subtracting the R_{RP} (high rcf) vector (intensity 15 mA m^{-1}; 309°, − 25°) from each of the 100–290 mT vectors. R_{RP} (low rcf) is the mean of the 15 vectors obtained by subtracting the ($R_{NP} - R_{RP}$ (high rcf)) vector from each of the NRM to 80 mT vectors. The statistics of the mean directions are R_{RP} (high rcf), $N = 6$, 309°, −25°, $\kappa = 63$, α_{95}, 8°; R_{RP} (low rcf), $N = 15$, 314°, −23°, $\kappa = 112$, $\alpha_{95} = 4°$; R_{NP}, $N = 10$, 113°, +35°, $\kappa = 105$, $\alpha_{95} = 5°$. The directions of reverse and normal polarities are only 15° away from anti-parallelism. A possible solution to the contribution of the R_{RP} and R_{NP} remanences is shown in dashed lines and discussed in the text.

directed. Detailed af treatment and vector analysis show that this remanence was not acquired in a different field but is simply the resultant of two almost oppositely directed remanences: a normal polarity remanence (R_{NP}) and a reverse polarity remanence (R_{RP}) (Fig. 7). Upon af treatment the direction moves gradually into the normal polarity group at 100–150 mT and for higher fields returns toward the reverse polarity. The shift in direction is in accordance with the observed remanence decay curve (Fig. 7, squares). The minimum at 30 mT corresponds to the mid-distance between the NRM direction and the site-mean direction, the latter being reached at the maximum (100 mT) where the R_{NP} is clearly predominant. The R_{NP} and the R_{RP} both have some rcf of very high coercivities since the direction of the remanence remaining after the 290 mT treatment lies between the two polarities. Although the rcf spectra of the two remanences overlap somewhat, it appears that most of the rcf spectrum of the R_{NP} is confined to the 100–150 mT interval while that of the R_{RP} is below 100 mT. To reconstruct the decay curves of the two constituent remanences (Fig. 7) certain assumptions had to be made: the R_{RP} had no rcf (no slope to the decay curve) over the 100–290 mT range and the R_{NP} had no rcf over the 0–100 mT range. Decomposition of an observed remanence into its constituents by vector analysis may be nonunique and constituent vectors obtained vectorially should not be expected to represent the true vectors but should be regarded as an approximation (Roy and Park 1974). Normally, certain criteria can be used to qualitatively assess the approximation. In this instance, a

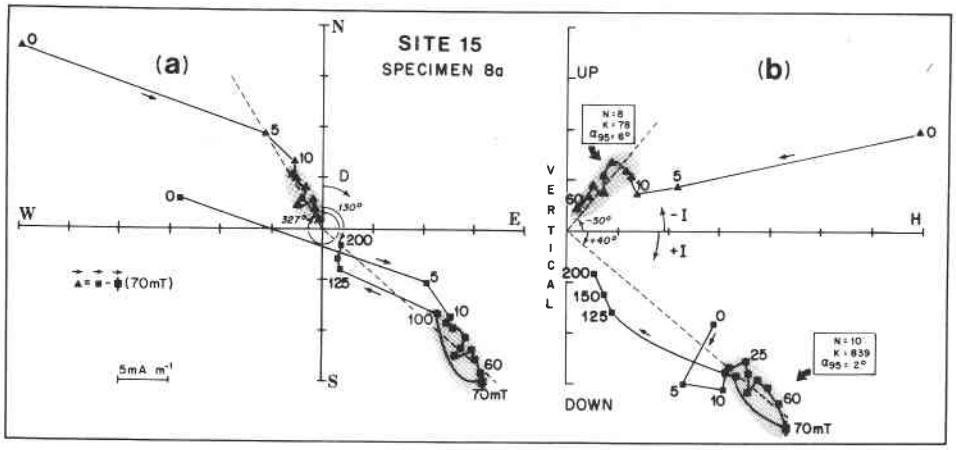


FIG. 8. Diagrams of the observed vector (squares) remaining after each af treatment step from site 15, specimen 8*a*. Only the termination of the vector is plotted. The horizontal component of the vector is shown in (*a*) and the total vector is shown in (*b*). The distance from the origin gives the length of the vector (intensity). In (*a*) the angular distance from the positive *x* axis (*N*) indicates the declination of the magnetic vector; in (*b*) the inclination is given by the angular distance from the H axis. These diagrams are particularly useful for illustrating vector analysis. The triangles show the vectors of the reverse polarity remanence (R_{RP}) obtained by successively subtracting the 70 mT vector from the 0–60 mT vectors. Dashed lines have been drawn through the mean directions showing the nearly anti-parallelism of the removed and observed vectors. Statistics are given for the vectors lying in the shaded area. It is noted that the dashed line drawn through the 15–100 mT observed vectors represents the sum of the R_{RP} and R_{NP} (normal polarity remanence) vectors. The R_{NP} can be obtained by subtracting the R_{RP} from the observed vectors. Since the vector ratio of R_{RP}/observed is small, the individual R_{NP} vector lies close (1–7° clockwise) to the corresponding observed vector, and those vectors have not been plotted to avoid cluttering. The mean direction of the 10 R_{NP} (15–100 mT) is 134°, 42° with $\kappa = 616$ and $\alpha_{95} = 2°$. The R_{RP} and R_{NP} are within 14° of anti-parallelism.

slight slope of the R_{NP} decay curve in the 0–100 mT range may be compensated for by an increase in the slope of the R_{RP} decay curve to obtain the observed remanence decay curve. However, any change in slope also affects the direction obtained by vector subtraction and consequently alters the dispersion among directions of successively subtracted vectors (following the method described in the Fig. 7 legend). The precision estimate, κ, can be used to evaluate the closeness of the approximation. By slightly altering this model, the mean directions do not change substantially; the precision estimates, however, do become smaller. This, and the fact that the R_{RP} and R_{NP} have very high κ and small α_{95} suggest that the model proposed represents a good approximation of the constituent remanences. The near anti-parallelism (within 15°) of the two directions leaves little doubt that the specimen carries a remanence of dual polarities only and that the NRM, although northerly directed, is not attributable to a later (Paleozoic?) remagnetization. All other results from dispersed NRMs are shown in Fig. 6. Great circles passing through the direction after treatment, the NRM direction, and the direction obtained from vector subtraction show the presence of two nearly opposite remanences.

The scatter of NRM directions at site 15 (Fig. 3*b*) is also most probably attributable to the same cause. As an example, for specimen 8*a*, vectorially subtracting the 70 mT vector from the vectors obtained after lower treatment results in the directions of the normal and reverse polarity remanences being within 14° of anti-parallelism (Fig. 8).

The cataclastic anorthosite from site 14 was subdivided into sites 14*a* (higher intensity) and 14*b* (lower intensity). The normal polarity remanence can be recognized at site 14*a* (Fig. 9*b*) although the inclination is appreciably lower (≈25°) than that found at the St-Urbain anorthosite sites (Table 1). The NRM directions at site 14*b* were scattered. Upon af treatment the vector in some cases moved towards the site-mean direction of 14*a* and then wandered about a more southerly direction, roughly along a great-circle path (Fig. 9*c*). The mean direction calculated for five specimens after treatment at 200 mT is 178°, +38°, $\kappa = 10$, $\alpha_{95} = 26°$. Upon thermal treatment the direction of four specimens moves to 193°, +04°, $\kappa = 42$, $\alpha_{95} = 14°$ at 500°C and then to 176°, +22°, $\kappa = 61$, $\alpha_{95} = 12°$ at 670°C (Fig. 9*b*). Again, this remanence can be explained as the resultant of unresolved reverse and normal polarity remanences. Although the af results are inconclusive, the four thermal results all show a definite trend (500–670°C) to-

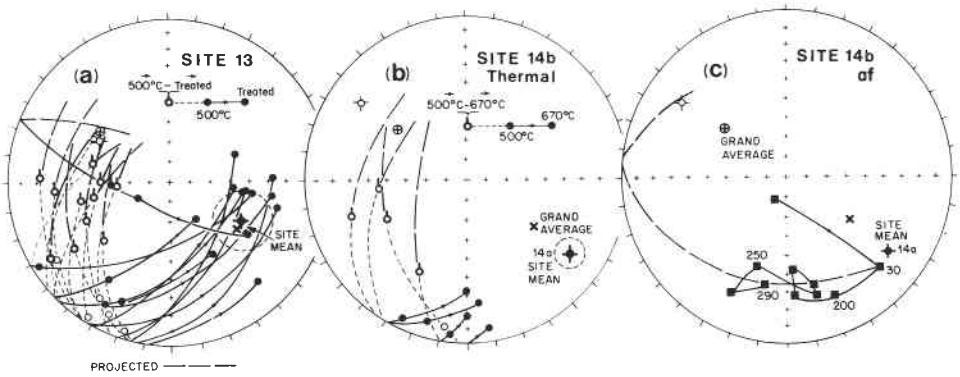


FIG. 9. Directions obtained after the indicated treatments. Directions obtained by vector subtraction and projection along great circles are shown. Polarities of both the site mean and grand average 3 (see text) are also shown. It is argued that the southerly directions may be the result of unresolved dual polarity remanences directed along the mean.

wards the normal polarity direction (Fig. 9*b*). Vector subtraction yields directions which approach the reverse polarity direction.

Shocked Rocks of the Central Uplift

The gabbroic anorthosite of site 13 corresponds to site 90 of Hargraves and Roy (1974) and the NRMs from both studies are southerly directed (≈220°) with a low inclination. Hargraves and Roy found little directional change after treatment at 50 mT. For the one specimen that we af treated, the vector remained in the 220° direction to 100 mT and then slowly progressed to higher declination (≈270°) between 100 and 175 mT, where the intensity became too low (0.4 m Am^{-1}) to be measured reliably. For the thermally treated specimens (Fig. 9*a*), the vectors wandered about this direction until 500°C, where the site-mean direction is 219°, +05° with $\kappa = 13$ and $\alpha_{95} = 22°$. However, upon further treatment at higher temperatures, the directions moved towards the direction obtained at the St-Urbain sites. Vector analysis indicates that the southerly directed remanence found at 500°C could easily be the resultant of normal and reverse polarity remanences as found at sites 8 and 15. The evidence for the reverse polarity at site 11 is obtained directly from measurements rather than by vector analysis. Upon thermal treatment, the directions of all specimens clearly move to a reverse polarity to yield a mean direction nearly opposite (within 13°) that of the mean normal polarity direction (Fig. 5).

Origin of Magnetization

Both the petrographic and paleomagnetic results indicate that the carrier of magnetization in the anorthosite is hematite or titano-hematite which was probably formed during and (or) shortly after cooling from above the Curie point of hematite (680°C) to produce a T-CRM (thermo-chemical remanent magnetization). During most of the magnetization process the ambient field was normally polarized, as indicated by the fact that at seven sites a single polarity remanence can be detected. Owing to the detection of a reverse polarity remanence at five sites, it is suggested that a field reversal occurred towards the beginning or the end of the magnetization process. A self-reversal mechanism could also be invoked to explain this latter remanence. Although such mechanisms are difficult to reject outright, they are even more difficult to defend, in this study at least. It is, for example, arduous to explain why at site 8, the self-reversal mechanism would be operative at the top (specimen 8*a*) and bottom (specimen 8*c*) but inoperative in the middle (specimen 8*b*) of the core (Fig. 6); likewise, in core 10, the bottom of the core (specimen 10*d*, Fig. 7) would have been subjected to this mechanism, whereas the top three specimens would have escaped. No petrographic or magnetic characteristics can be found to support such selectivity. Conversely, the slow process of CRM acquisition could easily occur partly before and partly after a field reversal. The magnetic characteristics of the resulting two nearly oppositely directed components of remanence would then be similar. This is in accordance with the results. The only requirement for this common explanation of all data is that the magnetization process that produced the thermochemical remanence lasted long enough (1–10 ka) to span field-reversal time scales.

At only two sites (13 and 14*b*) were different remanence directions detected. Although our vector analyses show that these two southerly direc-

tions can be adequately explained as the resultant of the normal and reverse polarity remanences discovered at sites 8 and 15, the possibility of a shock-induced component must be assessed. Hargraves and Roy (1974) considered that this remanence might have been produced at the time of formation of the structure (360 Ma), either parallel to the ambient field or in a field geometrically related to the direction of shock propagation. The first possibility cannot be entirely discounted since the direction of the field in North America during the Lower Paleozoic is not well known. This alternative is not favoured, however, as the most highly shocked samples (site 11), and, therefore, those presumably most susceptible to such an overprint, do not show this southerly direction. Sites 13 and 14*b* are roughly 14 km from the shock centre at Mont des Eboulements but along azimuths separated by approximately 85°. If the shock-induced remagnetization were related to the direction of shock propagation it would be expected that the remanence directions at these two sites would be similarly separated. As their declinations (Table 1) are within 5° (NRM) and 26° (500°C) of one another this second possibility also seems unlikely.

Pole Calculations

The virtual geomagnetic pole (VGP) derived from each site is listed in Table 1, except for sites 2 and 4. These latter sites were collected from small outcrops whose *in situ* positions were very much in doubt. Their directions of magnetization are considered unreliable for determination of VGP. They were sampled, nonetheless, to obtain more data about the magnetic characteristics of the anorthosite. The site-mean directions of the remaining 10 sites are given in Table 1. The reliability of the directions of sites 3 and 14*a* as paleofield direction indicators might be questioned. Site 3 is from an outcrop which lies near the inner margin of the downfaulted peripheral trough of the impact structure. Faults mapped in the anorthosite within one kilometre north and south of this site, and highly tilted limestones immediately to the east (Rondot 1972) give evidence of a variety of attitudes of the rocks in this region. The low inclination is likely due to tilting of this block to the west through faulting, rather than by updoming of the central uplift as proposed by Hargraves and Roy (1974) for their nearby sites 97 and 101. The reliability query at site 14*a* seems to be of a different nature. Considering that the direction of the initial magnetization could not be determined from site 14*b* because of dual polarity, one must then consider the possibility that the normal polarity remanence at site 14*a* has not been sufficiently isolated; if this is so, then an error in field direction determination would be introduced.

Because of these uncertainties, grand averages have been calculated using different combinations of sites, that is, including or excluding sites 3 and 14*a* (Table 1). Calculations using only the St-Urbain anorthosite sites (with or without site 3) are also given. The results show that the grand average direction and corresponding pole are not greatly affected in any combination. All directions and poles are within 7° of arc. However, in all cases where site 3 is excluded, values of κ are larger and those of α_{95} are smaller than in the corresponding cases where site 3 is included. Grand average 3 (124°, +48°; pole 156°E, 02°S), which includes all (except 3) sites, is regarded as the most accurate determination for this study. Perhaps the best estimate of the ancient field direction is given by grand average 7 (126°, +51°; pole 153°E, 03°S), calculated by giving unit weight to the nine sites of grand average 3 and the five sites of Hargraves and Roy (1974). Whichever average is used is unimportant for paleopole determination since both the directions and poles are within 4° of arc.

The pole falls on the polar path of Morris and Roy (1977) and Roy and Robertson (1978) at the 950 Ma calibration mark as determined from both Grenvillian and Interior-Laurentia rock units. This is in particularly good agreement with the estimated age of the St-Urbain anorthosite since it is believed to be slightly older than the cutting dykes dated at 890 Ma (Rose 1961).

Effect of Shock

No unequivocal evidence can be found at Charlevoix for a shock induced remagnetization at the time of impact (360 ± 25 Ma). At 10 of 12 sites from which meaningful results were obtained, the rocks carry a single vector remanence whose direction is consistent with the direction of the ambient field at the time of cooling. At only two sites (13 and 14) was a different direction (southerly) of magnetization detected. Although it may possibly represent a magnetization acquired along the ambient field at the time of impact, it is more probably the direction of the vector resultant of a dual polarity initial remanence.

Even at the centre of the impact structure (site 11) some of the initial remanence appears to have been preserved. The site-mean direction is oppositely polarized, but in good accord with the mean direction for the St-Urbain anorthosite sites. The

TABLE 2. Averages obtained from different combinations of Table 1 results

Key	N	D, I	R	κ	α_{95}	Pole (°E, °S)	$\delta p°, \delta m°$
(1) St-Urbain (site 3 exc.)	6	125°, +50°	5.87	39	11	154, 02	10, 15
(2) St-Urbain (site 3 inc.)	7	120°, +45°	6.59	15	16	161, 01	14, 20
(3) All sites (site 3 exc.)	9	124°, +48°	8.74	31	9	156, 02	08, 12
(4) All sites (site 3 inc.)	10	120°, +45°	9.46	17	12	160, 01	10, 15
(5) All sites (sites 3 and 14*a* exc.)	8	124°, +51°	7.85	46	8	155, 04	08, 11
(6) All sites (site 14*a* exc.)	9	120°, +47°	8.55	18	12	160, 03	11, 16
(7) Ave. 3 + HR sites	14	126°, +51°	13.62	34	7	153, 03	07, 10

NOTES: St-Urbain indicates that the average has been obtained from the St-Urbain anorthosite exclusively. Averages have been calculated by including (inc.) or excluding (exc.) sites 3 and 14*a*. Average 7 is the result obtained by using the 9 sites of average 3 and the 5 sites of Hargraves and Roy (1974). δp, δm are the semiaxes of the oval of confidence at $P = 0.05$. Averages 3 and 7 are considered the best estimates.

NRM intensity, however, is only approximately 6% of that in the St-Urbain pluton samples, and presumably has been reduced to this value by the impact. The disappearance of iron from what appear to be original hematite–ilmenite grains (Pl. 1, figs. 3–4), a phenomenon so far unexplained but undoubtedly a shock effect, must surely contribute to this overall intensity reduction. Such an explanation cannot be applied to sites 12 and 13 where the intensities are also anomalously low, but where no microscopic shock effects occur, nor would they have been expected at this distance from the impact centre. The weak NRMs at these two sites are likely a consequence of an initially low content of iron in magnetic phases as observed petrographically. However, the destruction or weakening of original remanence at equivalent distances from the shock centres at Slate Islands (Halls and Stesky 1978) and Lonar Lake craters (Cisowski 1975) should be kept in mind.

Why only the reversely polarized magnetization at site 11 is detectable is uncertain. Is it possible that the pre-impact remanence comprised a single component reversely magnetized, or could it be that the shock effects were limited to the carriers of the normal magnetization and the reverse component has survived? Remanent magnetization directions at site 11 (295°, −68°) and site 92 (326°, −68°) of Hargraves and Roy (1974), which lie within 0.5 km of each other, are similar. Our interpretation, based on the recognition of reversals in the unshocked rocks of the St-Urbain pluton, is that the direction at site 11 is an original remanence. Hargraves and Roy (1974), however, who were not able to detect reversals in the St-Urbain anorthosites, considered this direction to be either a shock-induced reversal or the original direction preserved in blocks highly tilted and rotated in the central uplift processes. Our interpretation requires that virtually no disorientation of the blocks occurred in the central uplift region. Evidence contrary to such a quiescent type of uplift is provided at the Gosses Bluff crater (Milton *et al.* 1972), where it is obvious from the stratigraphy that beds in the central region are highly disturbed. On the other hand, and in support of our contention, shatter cone orientations and paleomagnetic directions at Slate Islands (Halls and Stesky 1978) must also be interpreted in terms of virtually no disorientation in the central uplift.

We are left, apparently, with no shock-induced magnetization of anorthosites in the Charlevoix impact structure. The anorthosite of site 11 displays shock effects corresponding to pressures of about 25 GPa (1 GPa = 10 kbar) (Robertson 1975) and post-shock temperatures of approximately 250°C (Stöffler 1971). Although such temperatures lie well below the Curie point of hematite, Wasilewski (1973) believes that a partial thermoremanent magnetization can be produced by post-shock temperatures of this magnitude. Certainly the shock grade of the Slate Islands material which bears the shock magnetization, is significantly lower than in the anorthosites of site 11 (Grieve and Robertson 1976).

The absence of a shock-induced stable remanence may very possibly be related to the nature of the magnetic carrier. In the various instances referenced in the introduction where a shock remanence has been produced, whether experimentally or naturally, the ferromagnetic component is believed to have been magnetite or metallic iron or an alloy. In the two cases where hematite was the magnetic carrier, Meteor Crater and the Rainier Nuclear crater (Hargraves and Perkins 1969), no shock-induced remanence was established.

Conclusions

Detailed experimental work on samples of the St-Urbain anorthosite body indicates that all of the

remanence was acquired in a unique direction. This remanence has very high T_{UB} and rcf and its carrier is most probably the hematite phase observed in various textures usually in association with ilmenite. Vector analysis indicates that the remanence in some specimens is composed of two nearly oppositely directed components; the preferred explanation is that a field reversal occurred during the magnetization process. Since no other remanence could be detected, the time of acquisition was probably during and (or) shortly after cooling from above 700°C. Because a field reversal was recorded, the process of magnetization had to have been long enough to span field reversal time scales (1–10 ka). Depending on the rate of cooling, the remanence could be a T-CRM and (or) CRM. Because of the complexity of the acquisition of chemical and thermo-chemical remanences, the exact mode and time of acquisition cannot be determined. However, the reasonably good agreement of within- and between-site directions, indicates that the magnetization process was terminated before any appreciable apparent polar movement could take place. The direction of magnetization is taken to be a reasonable approximation of the ambient field direction at the time of cooling which is estimated to be slightly older than 890 Ma. The best estimate of the field direction is 126°, +51° which yields a pole at 153°E, 03°S, in good agreement with one of the best defined segments of the late Precambrian apparent polar path. A slightly different direction from site 3 is due to tilting of the outcrop block during post-crater adjustments which led to the downfaulted peripheral trough and uplifted central peak.

Some effects of shock on the magnetic remanence have been observed. At the most highly shocked site sampled within the central uplift (11) the stable remanence found in the unshocked rocks has been severely reduced or virtually destroyed. A less shocked central uplift site (13) shows both normal and reverse magnetization in the mean direction of the unshocked St-Urbain anorthosite, whereas the more highly shocked (11) contains only the reversal. Both polarities exist in at least three of the St-Urbain sites as well and are interpreted, therefore, as pre-impact and not attributable to the shock event. There is no convincing evidence that the removed remanence was replaced by a shock-induced or post-shock stable magnetization.

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